

North Ridge

Generated: 2026-09-09

SCIENTIFIC ACCURACY & INTEGRITY DISCLOSURE:

- **Coordinate System:** LOCAL_ARBITRARY (Monocular camera optical reference).
- **Scale Status:** RELATIVE_SCALE (No metric ground coordinates without validated calibration).
- **Georeferencing:** UNREFERENCED (WGS84 / GPS anchors not bound; GeoJSON disabled).
- **Reconstruction:** Authoritative sparse SfM; dense MVS was unexecuted due to hardware constraints and no synthetic dense points were fabricated.

1. Mission & Source Video Summary

Mission ID	north-ridge	Operator	Hexa Spark Inspection Team
Location / Sector	Operational Flight Zone	Mission Type	Single-Pass Inspection
Video Source	flight-video.mp4	Resolution	1920x1080
Frame Rate	25.0 FPS	Duration / Frames	5.0 s (125 frames)

2. AI Detection & Tracking Performance

Detector Model	aeromesh_yolo (aeromesh-visdrone)	Tracker	Ultralytics persistent ByteTrack
Total Detections	17	Unique Tracks	17
Confidence Stats	Mean: 0.000 Min: 0.000 Max: 0.000	Sampling Rate	2.0 FPS (61 frames)
Detections by Class	car: 15, truck: 1, person: 1	Tracks by Class	car: 15, truck: 1, person: 1

3. 3D Photogrammetry & Surface Reconstruction

SfM Camera Model	SIMPLE_PINHOLE	Registered Cameras	11 / 11 (100%)
Sparse Points	5,252	Mean Reprojection Error	0.9785 px
Surface Mesh Status	AVAILABLE (camera_poisson_trimmed)	Mesh Complexity	0 vertices 0 faces
Dense Reconstruction	AVAILABLE (0 synthetic points)	Coordinate Framework	LOCAL_ARBITRARY / RELATIVE_SCALE

4. AI-to-3D Multi-View Spatial Fusion

Authoritative 2D Tracks	85	Mean Reproj Error	0.000 px (threshold: 25.0 px)
Tracks Evaluated	85	Acceptance Rate	100.0%
Association Status	VALID: 49 LOW_CONF: 36 INSUFFICIENT: 0 REJECTED: 0	Motion States	STATIC: 79 MOVING: 6

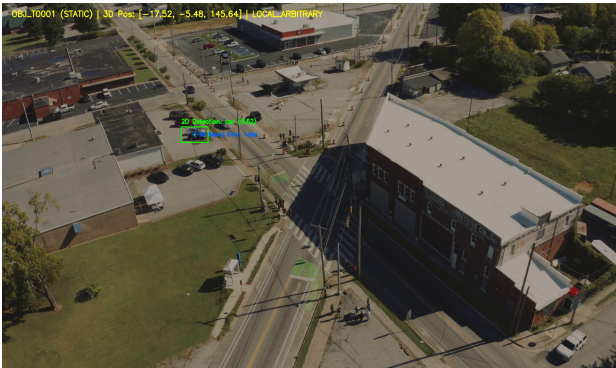
Object ID	Track	Class	Motion	Status	Position (X, Y, Z)	Reproj (px)
OBJ_T0001	T0001	van	STATIC	VALID	[-1.32, 1.15, 9.91]	12.59
OBJ_T0002	T0002	van	STATIC	VALID	[-1.26, 1.79, 8.12]	2.79
OBJ_T0003	T0003	van	MOVING	VALID	[-1.82, -2.03, 19.32]	23.85
OBJ_T0004	T0004	van	STATIC	LOW_CONFIDENCE	[-0.75, 2.77, 5.74]	0.06
OBJ_T0005	T0005	van	STATIC	VALID	[-0.88, 1.60, 8.38]	13.67
OBJ_T0006	T0006	van	STATIC	VALID	[-1.00, 1.09, 10.42]	13.79

5. Geometric Measurements & Metric Scale Calibration

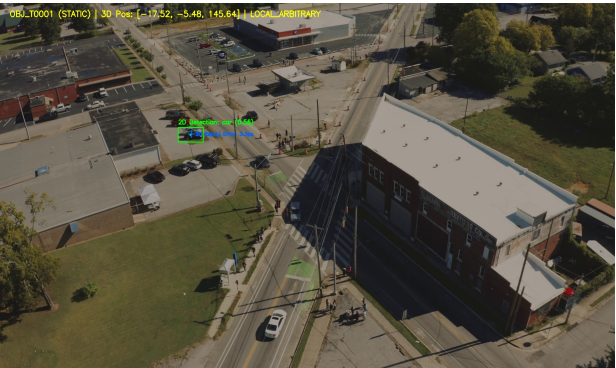
Calibration ID	None	Method	UNREFERENCED
Scale Factor	1.00000 m/unit	Known Baseline	N/A relative_units (Confidence: 0.00)

6. Visual Evidence & Reprojection Overlays

The following imagery represents authoritative multi-view observations reprojected onto the 3D model with 2D bounding boxes:



Overlay: overlay_OBJ_T0001_frame_00000.jpg



Overlay: overlay_OBJ_T0001_frame_00001.jpg

7. Comprehensive Mission Limitations & Disclosures

• LOCAL_ARBITRARY: Reconstruction coordinates are arbitrary relative units, not true meters or GPS coordinates.
• RELATIVE_SCALE: Monocular video SfM is scale-ambiguous; metric distances are strictly valid only where verified ground scale calibration is applied.
• UNREFERENCED: Scene is unreferenced against EPSG/WGS84. GeoJSON geographic coordinates (latitude/longitude) are unavailable.
• DENSE_MVS_UNAVAILABLE: Dense stereo reconstruction requires CUDA or HIP hardware. Sparse SfM is preserved as authoritative geometry without synthetic point fabrication.

- **UNOBSERVED_SURFACES:** Unobserved, occluded, or low-parallax areas are preserved as unobserved rather than interpolated as synthetic geometry.